



COBOTS WORK ALONGSIDE HUMANS

To Do Dull, Dirty, Dangerous, Dear Jobs



Human workers, who once saw industrial robots as a threat to their jobs, are now quite happy to share the space with the current generation of cobots, as these are proving to be truly cooperative, collaborative, and congenial co-workers

Venkatesh Gurappa (Doc Venki) with one of his robots (Courtesy: Bangalore Robotics)



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Some years ago, a hammer fell into a high-pressure steam pipe at Tata Power's plant in Mundra, Gujarat, stalling the plant's operations. It was a pretty complex situation, as the 90-degree bend of the pipe made it almost impossible to see where the hammer was stuck. Interestingly, the plant sought the help of Gridbots Technologies, an Ahmedabad based robotics startup. The Gridbots team made some modifications to the gripper of their Stinger pipeline robot and managed to get the hammer out in a five-hour operation, at a cost of ₹1,600,000. Compare that with all other alternatives, which would have taken several days and cost the company tens of billions of rupees in expenses and losses, and you will understand why a lot of robotics enthusiasts remember and recall this story even after so many years.

That indeed is what robots are meant to do. In the robotics world, this is known as the 4D principle—let robots do tasks that are dull, dirty, dangerous, or dear. That is, work that is repetitive, unhygienic, risky, or costly (where one cannot afford to err). For this purpose, robots have been used in heavy industries like automotive, and dangerous conditions like mines, for a long time now.

However, the robots of yore were generally expensive, huge, heavy, and often dangerous. They had to be confined to work-cells, away from human workers. Most robots were also difficult to program, and hence used for specific, unchanging tasks. This put them out of the reach of small and medium sized industries characterised by high-mix, low-volume jobs. On top of everything else, human workers often looked at these intimidating robots as the bad guys who had come to steal their lunch!